

系統程式設計

Lab3 Demo Answers

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I. How do you direct the output file to '/dev/null'?
By using the open system call with "O_WRONLY" flag on path "/dev/null".

II. The table of Part3
Does the bigger bufsize make IO faster?
Yes, by increasing buffer size the IO times decrease, but only to a certain degree

III. The table of Part4
What happened? Why?
The second time running "time ./bufsize4096 < 512mb_file2", the real time was lower than the first run.

Due to linux page cache, in the first run the data is being copied from the disk to RAM, but on the second run the OS realizes that the data's already in RAM hence improving real time speed.

IV. The table of Part5
What happened? Why?
The real time increased a lot and the caching from Part4 didn't happen.

When fsync is in the loop, it forces the OS to pause and wait till the data is stored on the physical disk instead of telling the program that the write is done when the data's stored in RAM.